

LaTeX Tutorial

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1 Introduction

The eigenvectors and eigenvalues of a linear transformation serve to characterize it, and so they play important roles in all areas where linear algebra is applied, from geology to quantum mechanics. In particular, it is often the case that a system is represented by a linear transformation whose outputs are fed as inputs to the same transformation (feedback). In such an application, the largest eigenvalue is of particular importance, because it governs the long-term behavior of the system after many applications of the linear transformation, and the associated eigenvector is the steady state of the system. See the image for shear mapping Image 1

This information is based on data from Wikipedia [2]

1.1 Overview

Eigenvalues and eigenvectors feature prominently in the analysis of linear transformations. The prefix eigen- is adopted from the German eigen (cognate with the English word own) for 'proper', 'characteristic', 'own'. Originally used to study principal axes of the rotational motion of rigid bodies, eigenvalues and eigenvectors have a wide range of applications, for example in stability analysis, vibration analysis, atomic orbitals, facial recognition, and matrix diagonalization.

This information is based on data from GeeksforGeeks [1]

2 Eigenvalues and eigenvectors of a matrix

Eigenvalues and eigenvectors are often introduced to students in the context of linear algebra courses focused on matrices. Furthermore, linear transformations over a finite-dimensional vector space can be represented using matrices, which is especially common in numerical and computational applications.

$$X = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 20 \\ -60 \\ 80 \end{bmatrix}$$

These vectors are said to be scalar multiples of each other, or parallel, or collinear, if there is $Y = \lambda X$ in this case $\lambda = -20$

See the Table 1 for Eigenvalues of geometric transformations

2.1 History

- Eigenvalues are often introduced in the context of linear algebra or matrix theory. Historically, however, they arose in the study of quadratic forms and differential equations.

- In the 18th century, Leonhard Euler studied the rotational motion of a rigid body, and discovered the importance of the principal axes. Joseph-Louis Lagrange realized that the principal axes are the eigenvectors of the inertia matrix.
- In the early 19th century, Augustin-Louis Cauchy saw how their work could be used to classify the quadric surfaces, and generalized it to arbitrary dimensions. Cauchy also coined the term *racine caractéristique* (characteristic root), for what is now called eigenvalue; his term survives in characteristic equation.

2.2 Topics to Discuss

1. Eigenvalues and characteristic polynomial
2. Spectrum of a matrix
3. Algebraic multiplicity

	Scaling	Unequal Scaling	Rotation
Matrix	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$	$\begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
Characteristic polynomial	$(\lambda - k)^2$	$(\lambda - k_1)(\lambda - k_2)$	$\lambda^2 - 2\cos(\theta)\lambda + 1$
Algebraic mult.,	$\mu_1 = 2$	$\mu_1 = 1 \mu_2 = 1$	$\mu_1 = 1 \mu_2 = 1$

Table 1: Eigenvalues of geometric transformations

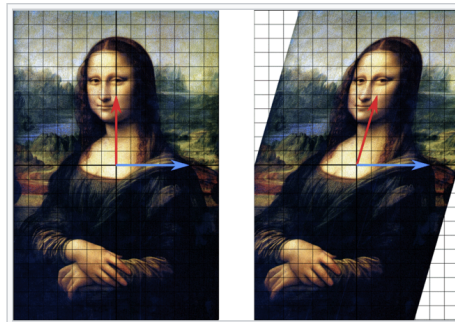


Figure 1: Shear Mapping of an image

References

- [1] GeeksforGeeks contributors. Eigenvalues and eigenvectors, 2026. [Online; accessed 3-July-2026].
- [2] Wikipedia contributors. Eigenvalues and eigenvectors, 2026. [Online; accessed 3-July-2026].